UNIVERSITY OF ABERDEEN

CELEBRATING 525 YEARS 1495 – 2020

ABERDEEN 2040

Association Rule Mining – 2

Data Mining & Visualisation

Lecture 22

Association Rule Mining: Step-by-Step

Association Rule Mining: Step-by-Step

Let’s walk through an example of the process.

Consider the following transactions involving five items. You have been asked to produce association rules for these items using the Apriori algorithm:

Transaction ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

i) Using a minimum support of 0.75, generate the frequent itemsets for the above data showing clearly the application of Apriori principle in pruning infrequent itemsets.

ii) Using a minimum confidence of 0.75, generate the association rules generated from the frequent itemsets computed in (i) showing clearly the application of Apriori principle in pruning low confidence rules.

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

i) **Using a minimum support of 0.75, generate the frequent itemsets for the above data showing clearly the application of Apriori principle in pruning infrequent itemsets.**

ii) Using a minimum confidence of 0.75, generate the association rules generated from the frequent itemsets computed in (i) showing clearly the application of Apriori principle in pruning low confidence rules.

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant ←
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant ←
4	Apple, Carrot, Durian, Eggplant ←

minsup = 0.75

First step, let's identify *frequent* 1-itemsets

$$Support = \frac{X \cap Y.count}{n}$$

{Apple}: 3 / 4 = 0.75 ✅

{Broccoli}:

{Carrot}:

{Durian}:

{Eggplant}:

We see that Apple appears in 3 transactions (X.count)

We also see that we have 4 total transactions (n)

Therefore, the support of our {Apple} 1-itemset is 0.75.

Since Support ≥ minsup, {Apple} is a frequent 1-itemset.

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

minsup = 0.75

First step, let's identify *frequent* 1-itemsets

$$Support = \frac{X \cap Y.count}{n}$$

{Apple}: 3 / 4 = 0.75 ✅
{Broccoli}: 3 / 4 = 0.75 ✅
{Carrot}: 2 / 4 = 0.5 ❌
{Durian}: 4 / 4 = 1.0 ✅
{Eggplant}: 3 / 4 = 0.75 ✅

After checking the support of each of our 1-itemsets, we can see that 4 of our itemsets are frequent.

Carrot does not reach the minimum level of support. Therefore, we use the Apriori principle to prune it.

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

minsup = 0.75

So far, our *frequent* itemsets are:

{Apple}, {Broccoli}, {Durian}, {Eggplant}

From these, our *candidate* 2- itemsets are:

{Apple, Broccoli}, {Apple, Durian}, {Apple, Eggplant},
{Broccoli, Durian}, {Broccoli, Eggplant}, {Durian, Eggplant}

Note that we do not make any candidate 2-itemsets using Carrot, since the Apriori principle tells us that all supersets containing Carrot will also be infrequent!

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

minsup = 0.75

Next, lets go back and count the number of transactions that include our *candidate* 2-itemsets:

{A, B}: 2 / 4 = 0.5 ❌
{A, D}: 3 / 4 = 0.75 ✅
{A, E}: 3 / 4 = 0.75 ✅
{B, D}: 3 / 4 = 0.75 ✅
{B, E}: 2 / 4 = 0.5 ❌
{D, E}: 3 / 4 = 0.75 ✅

After checking the support of each of our 2-itemsets, we can see that 4 of our itemsets are frequent.

{Apple, Broccoli} and {Broccoli, Eggplant} do not reach the minimum level of support. Therefore, we use the Apriori principle to prune them.

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

minsup = 0.75

So far, our *frequent* itemsets are:

{A}, {B}, {D}, {E}, {A, D}, {A, E}, {B, D}, {D, E}

From these, our *candidate* 3- itemsets are:

{A, D, E}

From our frequent 2-itemsets, we can only produce one candidate 3-itemset, since the Apriori principle tells us that any other combinations will be infrequent!

E.g. since {A, B} is infrequent, we know that {A, B, D} will also be infrequent.

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

minsup = 0.75

So, lets go back and count the number of transactions that include our *candidate* 3-itemset:

{A, D, E}: 3 / 4 = 0.75 ✅

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

minsup = 0.75

So far, our *frequent* itemsets are:

{A}, {B}, {D}, {E}, {A, D}, {A, E}, {B, D}, {D, E}, {A, D, E}

However, since we only have one frequent 3- itemset, we cannot produce any candidate 4-itemsets.

Therefore, we are done finding frequent itemsets!

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

i) Using a minimum support of 0.75, generate the frequent itemsets for the above data showing clearly the application of Apriori principle in pruning infrequent itemsets.

ii) Using a minimum confidence of 0.75, generate the association rules generated from the frequent itemsets computed in (i) showing clearly the application of Apriori principle in pruning low confidence rules.

Example Question

ID	Item List
1	Apple, Broccoli, Durian, Eggplant
2	Broccoli, Carrot, Durian
3	Apple, Broccoli, Durian, Eggplant
4	Apple, Carrot, Durian, Eggplant

minsup = 0.75

So far, our *frequent* itemsets are:

{A}, {B}, {D}, {E}, {A, D}, {A, E}, {B, D}, {D, E}, {A, D, E}

Let's now generate some *candidate* association rules.

Example Question

Itemset	Support
A	3 / 4 = 0.75
B	3 / 4 = 0.75
D	4 / 4 = 1
E	3 / 4 = 0.75
A, D	3 / 4 = 0.75
A, E	3 / 4 = 0.75
B, D	3 / 4 = 0.75
D, E	3 / 4 = 0.75
A, D, E	3 / 4 = 0.75

Minconf = 0.75

So far, our *frequent* itemsets are:

{A}, {B}, {D}, {E}, {A, D}, {A, E}, {B, D}, {D, E}, {A, D, E}

Our *candidate* association rules are:

$\{A \Longrightarrow D\}$	$\{D \Longrightarrow B\}$	$\{D \Longrightarrow A, E\}$
$\{D \Longrightarrow A\}$	$\{D \Longrightarrow E\}$	$\{A, E \Longrightarrow D\}$
$\{A \Longrightarrow E\}$	$\{E \Longrightarrow D\}$	$\{E \Longrightarrow A, D\}$
$\{E \Longrightarrow A\}$	$\{A \Longrightarrow D, E\}$	$\{A, D \Longrightarrow E\}$
$\{B \Longrightarrow D\}$	$\{D, E \Longrightarrow A\}$	

Example Question

Itemset	Support
A	3 / 4 = 0.75
B	3 / 4 = 0.75
D	4 / 4 = 1
E	3 / 4 = 0.75
A, D	3 / 4 = 0.75
A, E	3 / 4 = 0.75
B, D	3 / 4 = 0.75
D, E	3 / 4 = 0.75
A, D, E	3 / 4 = 0.75

Minconf = 0.75

Our *candidate* association rules are:

$\{A \Longrightarrow D\}$	0.75 / 0.75 = 1
$\{D \Longrightarrow A\}$	0.75 / 1 = 0.75
$\{A \Longrightarrow E\}$	0.75 / 0.75 = 1
$\{E \Longrightarrow A\}$	0.75 / 0.75 = 1
$\{B \Longrightarrow D\}$	0.75 / 0.75 = 1
$\{D \Longrightarrow B\}$	0.75 / 1 = 0.75
$\{D \Longrightarrow E\}$	0.75 / 1 = 0.75
$\{E \Longrightarrow D\}$	0.75 / 0.75 = 1
$\{A \Longrightarrow D, E\}$	0.75 / 0.75 = 1
$\{D, E \Longrightarrow A\}$	0.75 / 0.75 = 1
$\{D \Longrightarrow A, E\}$	0.75 / 1 = 0.75
$\{A, E \Longrightarrow D\}$	0.75 / 0.75 = 1
$\{E \Longrightarrow A, D\}$	0.75 / 0.75 = 1
$\{A, D \Longrightarrow E\}$	0.75 / 0.75 = 1

$$Confidence(X \Rightarrow Y) = \frac{Supp(X \cap Y)}{Supp(X)}$$

Example Question

In this case, all of our *candidate* association rules are valid, with confidence ≥ minconf.

Our *final* association rules are:

$\{A \Longrightarrow D\}$
$\{D \Longrightarrow A\}$
$\{A \Longrightarrow E\}$
$\{E \Longrightarrow A\}$
$\{B \Longrightarrow D\}$

$\{D \Longrightarrow B\}$
$\{D \Longrightarrow E\}$
$\{E \Longrightarrow D\}$
$\{A \Longrightarrow D, E\}$
$\{D, E \Longrightarrow A\}$

$\{D \Longrightarrow A, E\}$
$\{A, E \Longrightarrow D\}$
$\{E \Longrightarrow A, D\}$
$\{A, D \Longrightarrow E\}$

Itemset	Support
A	3 / 4 = 0.75
B	3 / 4 = 0.75
D	4 / 4 = 1
E	3 / 4 = 0.75
A, D	3 / 4 = 0.75
A, E	3 / 4 = 0.75
B, D	3 / 4 = 0.75
D, E	3 / 4 = 0.75
A, D, E	3 / 4 = 0.75

Minconf = 0.75

Association Rule Mining: Overview

Association Rule Mining

The space of all association rules is exponential: $O(2^m)$, where m is the number of items in I.

Our mining can exploit the sparseness of data, and high minsup and minconf values.

However, the process can still produce a huge number of rules, depending on the dataset.

Association Rule Mining

Nevertheless, the Apriori algorithm is very fast, and can scale up to large datasets.

It makes, at most, k passes over the data.

Typically, k is bounded (e.g. 10).

Problems with Association Rule Mining

With a single minsup value, we assume that all items within the dataset are of the same nature, and/or have similar frequencies.

However, in many applications, some items appear very frequently in the data, whereas other items rarely appear.

In a supermarket, for example, people buy bread and milk much more frequently than they buy food processors and cooking pots.

The Rare Item Problem

If the frequency of items varies a great deal, then we will encounter two problems:

- If minsup is set too high, then the rules that involve rare items will never be found
- If minsup is set too low, this can cause ‘combinatorial explosion’, since frequent items will be associated with each other in every possible way

The Multiple Minsup Model

To address this, we can introduce a model giving more flexibility to our minsup value.

Here, the minimum support of a rule is expressed in terms of minimum item supports (MIS) of the items that appear in the rule.

Each item can have a minimum item support.

By providing different MIS values for different items, we effectively express different support requirements for different rules.

The Multiple Minsup Model

With the Multiple Minsup Model, the minsup of a rule R is <u>the lowest MIS value of all of the items within the rule</u>.

E.g. Consider the following items: **bread, shoes, clothes**

With MIS values: MIS(bread) = 2% MIS(shoes) = 0.1% MIS(clothes) = 0.2%

The following rule **doesn't** satisfy its minsup: clothes $\Longrightarrow$ bread [sup=0.15%, conf =70%]

The following rule **does** satisfy its minsup: clothes $\Longrightarrow$ shoes [sup=0.15%, conf =70%]

The Apriori Principle with Multiple Minsup

In the Multiple Minsup Model, the Apriori principle no longer holds!

E.g., Consider items 1, 2, 3 and 4. Their minimum item supports are:

MIS(1) = 10% MIS(2) = 20% MIS(3) = 5% MIS(4) = 6%

{1, 2} with support 9% is infrequent, but {1, 2, 3} and {1, 2, 4} could both be frequent.

Mining Class Association Rules (CAR)

Note that in ‘normal’ association rule mining does not have any target.

It finds all possible rules that exist within the data, i.e., any item can appear as a consequent or a condition of a rule.

However, in some applications, the user is interested in some targets.

E.g, the user has a set of text documents from some known topics. He/she wants to find out what words are associated or correlated with each topic.

Problem Definition

Let T be a transaction dataset consisting of n transactions.

Each transaction is also labeled with a class y.

Let I be the set of all items in T, and Y be the set of all class labels.

A **class association rule** (**CAR**) is an implication of the form

$$x \Longrightarrow y, \quad \text{where} \quad x \subseteq I, \quad \text{and} \quad y \in Y.$$

The definitions of **support** and **confidence** are the same as those for normal association rules.

Example

Imagine a text document data set:

doc 1:	Student, Teach, School	: Education
doc 2:	Student, School	: Education
doc 3:	Teach, School, City, Game	: Education
doc 4:	Baseball, Basketball	: Sport
doc 5:	Basketball, Player, Spectator	: Sport
doc 6:	Baseball, Coach, Game, Team	: Sport
doc 7:	Basketball, Team, City, Game	: Sport

Let *minsup* = 20% and *minconf* = 60%.

Two examples of class association rules are:

Student, School $\Longrightarrow$ Education:

[sup= 2/7, conf = 2/2]

game $\Longrightarrow$ Sport:

[sup= 2/7, conf = 2/3]

Summary

Association rule mining is a technique that has been extensively studied within the data mining community.

There are many different algorithms and model variations that can be used across a wide variety of situations and contexts.